

Ecological forecasting in R

Lecture 1: introduction to time series models

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0900–1200 CET Monday 24th March, 2025

Welcome

About me

UQ Amplify Early Career Fellow

The University of Queensland
School of Veterinary Science
Located in Gatton, Australia

Expertise in:

Quantitative ecology

Molecular genetics

Multivariate time series modelling



Workflow

Press the "o" key on your keyboard to navigate among slides

Access the [tutorial html here](#)

Download the data objects and exercise R script from the html file

Complete exercises and use Slack to ask questions

Relevant open-source materials include:

[Forecasting Principles and Practice](#)

[Applied Time Series Analysis](#)

[Ecological Forecasting & Dynamics Course](#)

[How to interpret nonlinear effects from GAMs](#)

This lecture's topics

Why forecast?

Why are time series difficult?

Visualizing time series

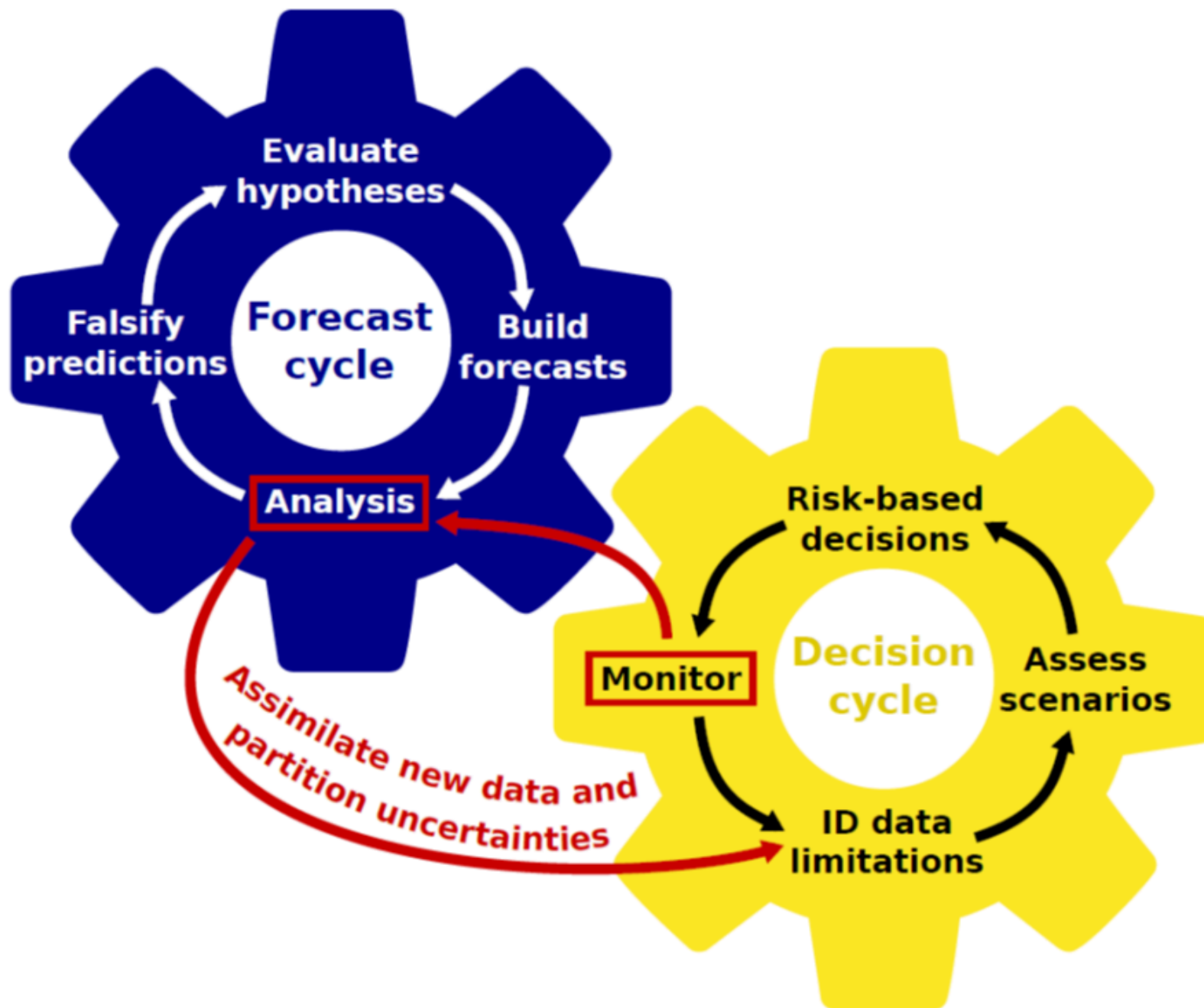
Common time series models

Why they fail in ecology

Why forecast?

“Because all decision making is based on what will happen in the future, either under the status quo or different decision alternatives, decision making ultimately depends on forecasts”

Dietze et al. 2018





Where is forecasting used?

Fisheries stocks, landings and bycatch risks

Coral bleaching and algal bloom risks

Carbon stocks

Wildlife population dynamics

Many other examples

NOAA Coastwatch's EcoCast

Tell fishers where to avoid bycatch

Harnesses up-to-date information for ecological models:

- Fisheries bycatch data

- Satellite observations

- Oceanography products

Builds distribution models and dynamically updates maps





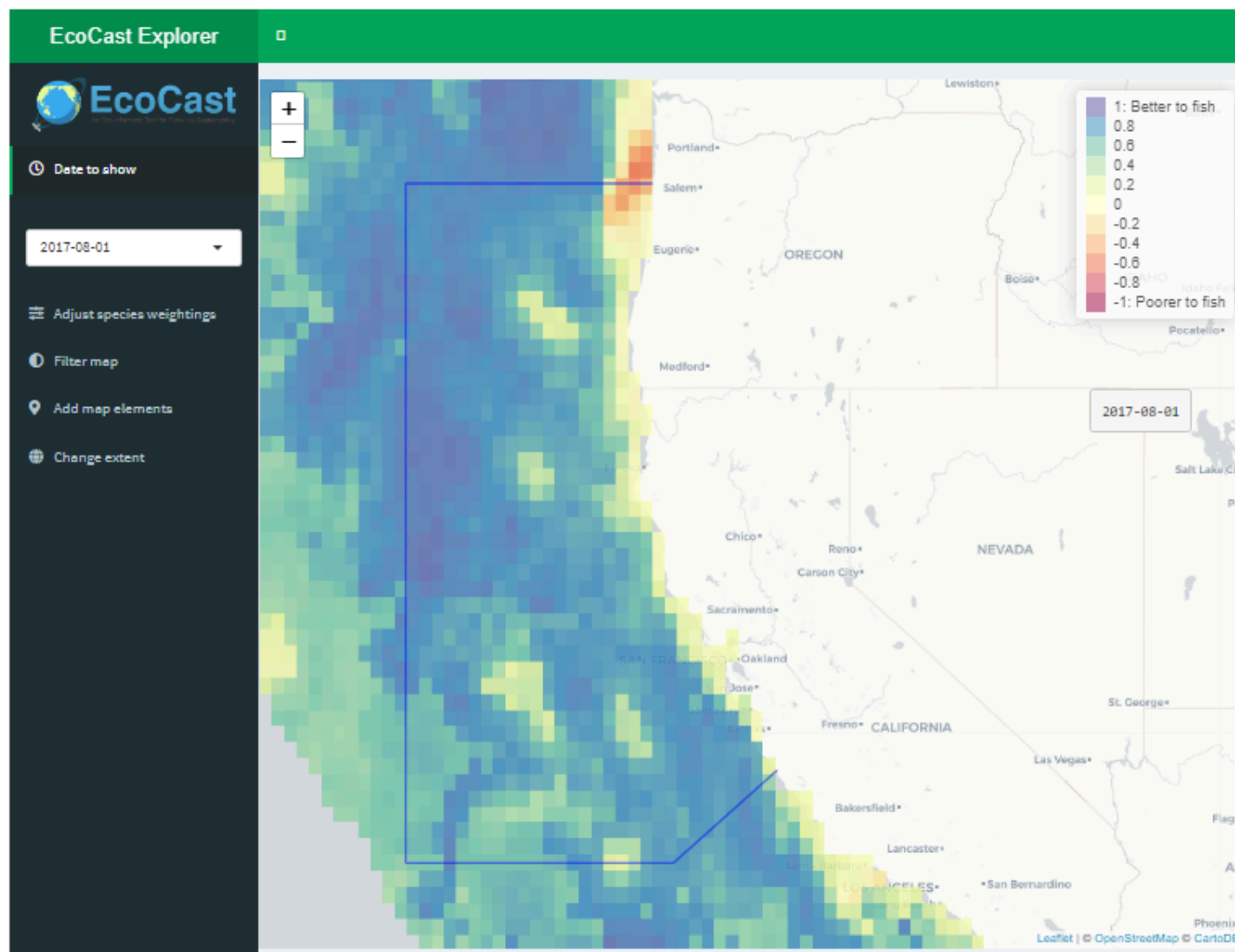
EcoCast Map

EcoCast Explorer

About EcoCast

EcoCast Explorer

Information and Instructions



Portal Project's Portalcast

Predict rodent abundance up to one year ahead

Harnesses up-to-date information for ecological models:

- Rodent captures from baited traps

- Satellite observations

Builds time series models and dynamically update forecasts



Species

Dataset

Model

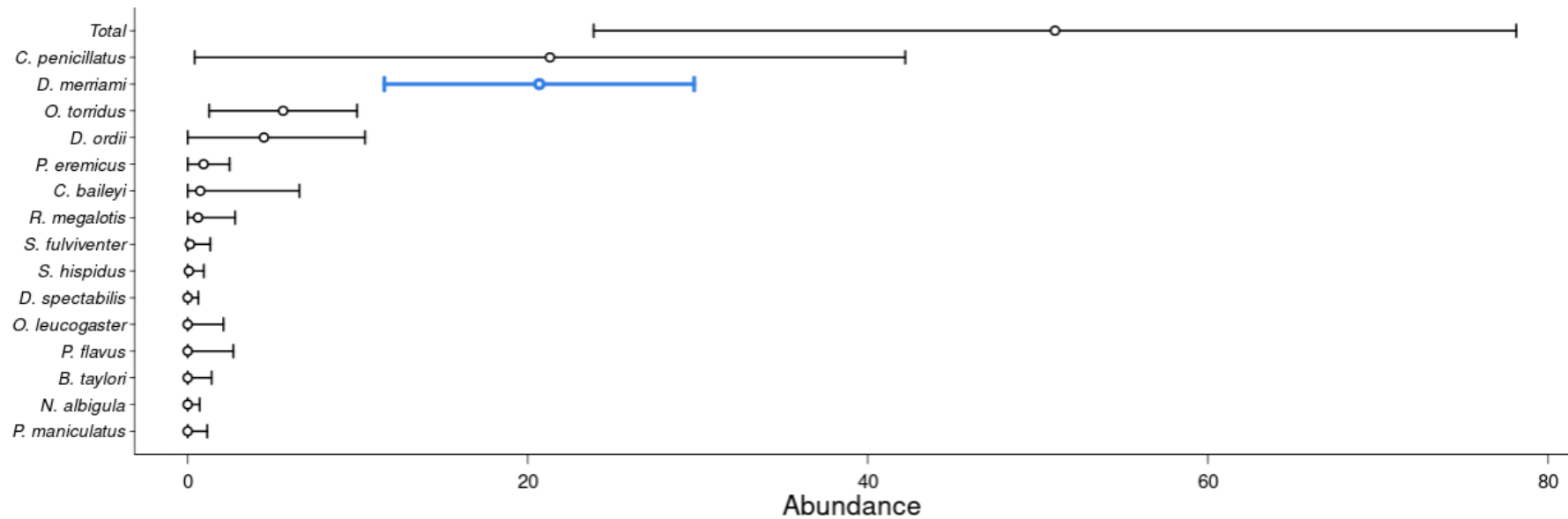
Origin Newmoon

Dipodomys merriami

controls

AutoARIMA

567



Why are time
series
difficult?

Some challenges of time series

Temporal autocorrelation

Lagged effects

Non-Gaussian data and missing observations

Measurement error

Time-varying effects

Nonlinearities

Multi-series clustering

Let's focus on these for now

Temporal autocorrelation

Lagged effects

Non-Gaussian data and missing observations

Measurement error

Time-varying effects

Nonlinearities

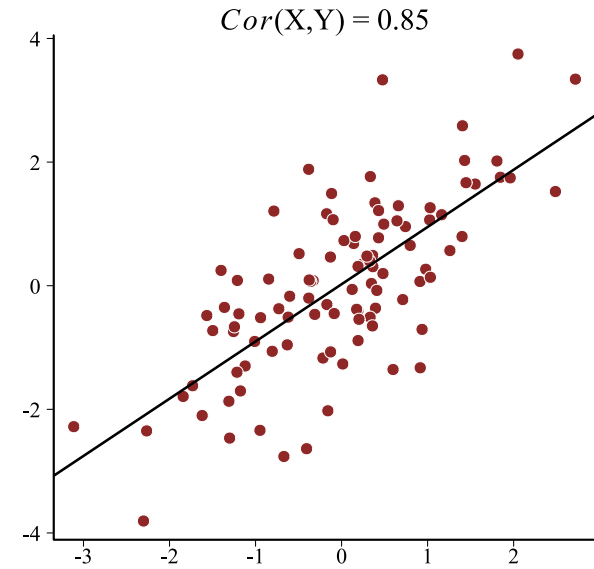
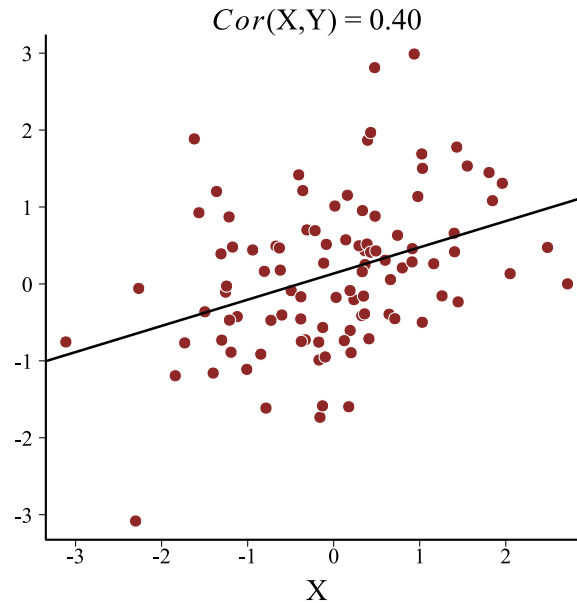
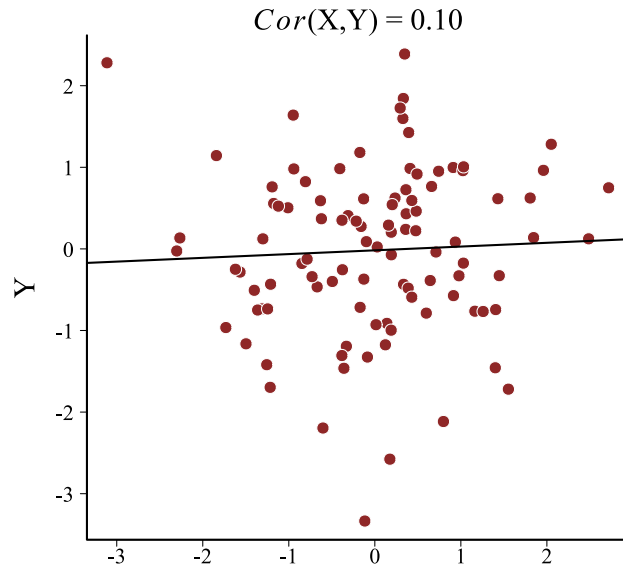
Multi-series clustering

What is temporal autocorrelation?

Values at current time *correlated with past values*

$$Cor(Y_t, Y_{t-lag}) \neq 0$$

Refresher: what is correlation?



Correlation assumes a *linear* relationship among two variables

What is temporal autocorrelation?

Values at current time *correlated with past values*

$$Cor(Y_t, Y_{t-lag}) \neq 0$$

We can estimate the correlation (β) with linear regression

$$\mathbf{Y}_t \sim \text{Normal}(\alpha + \beta * \mathbf{Y}_{t-lag}, \sigma)$$

What is temporal autocorrelation?

Values at current time *correlated with past values*

$$\text{Cor}(Y_t, Y_{t-lag}) \neq 0$$

We can estimate the correlation (β) with linear regression

$$\mathbf{Y}_t \sim \text{Normal}(\alpha + \beta * \mathbf{Y}_{t-lag}, \sigma)$$

Generalize to state that current value of a series (at time t) is *a function* of it's own past values (at time $t - lag$)

$$\mathbf{Y}_t \sim f(\mathbf{Y}_{t-lag})$$

A positively autocorrelated series

Code	Model	Plot
------	-------	------

```
# set seed for reproducibility
set.seed(1111)

# number of timepoints
T ← 100

# use arima.sim to simulate from an AR(1) model
series ← arima.sim(model = list(ar = 0.8), n = T, sd = 1)

# plot the time series as a line
plot(series, type = 'l', bty = 'n', lwd = 2,
      col = 'darkred', ylab = 'Y', xlab = 'Time')
```

A negatively autocorrelated series

Code	Model	Plot
------	-------	------

```
# set seed for reproducibility
set.seed(1111)

# number of timepoints
T ← 100

# use arima.sim to simulate from an AR(1) model
series ← arima.sim(model = list(ar = -0.8), n = T, sd = 1)

# plot the time series as a line
plot(series, type = 'l', bty = 'n', lwd = 2,
      col = 'darkred', ylab = 'Y', xlab = 'Time')
```

Correlations *over* >1 lag

Can include multiple lags of the same predictor variable (the response in this case)

$$\mathbf{Y}_t \sim f(\mathbf{Y}_{t-1}, \mathbf{Y}_{t-2}, \mathbf{Y}_{t-3})$$

Lagged effects of predictors

External conditions (eg temperature, humidity, landcover) can also influence what happens to a series at later timepoints

$$\mathbf{Y}_t \sim f(\mathbf{Y}_{t-lag}, \mathbf{X}_{t-lag})$$

Where:

$\mathbf{X}_t$ is the matrix of predictor values at time t

Many series show complex correlation structures; they can also show other temporal patterns

Seasonality

Many time series show
repeated periodic cycles

Breeding seasons

Migration

Green-ups / green-downs

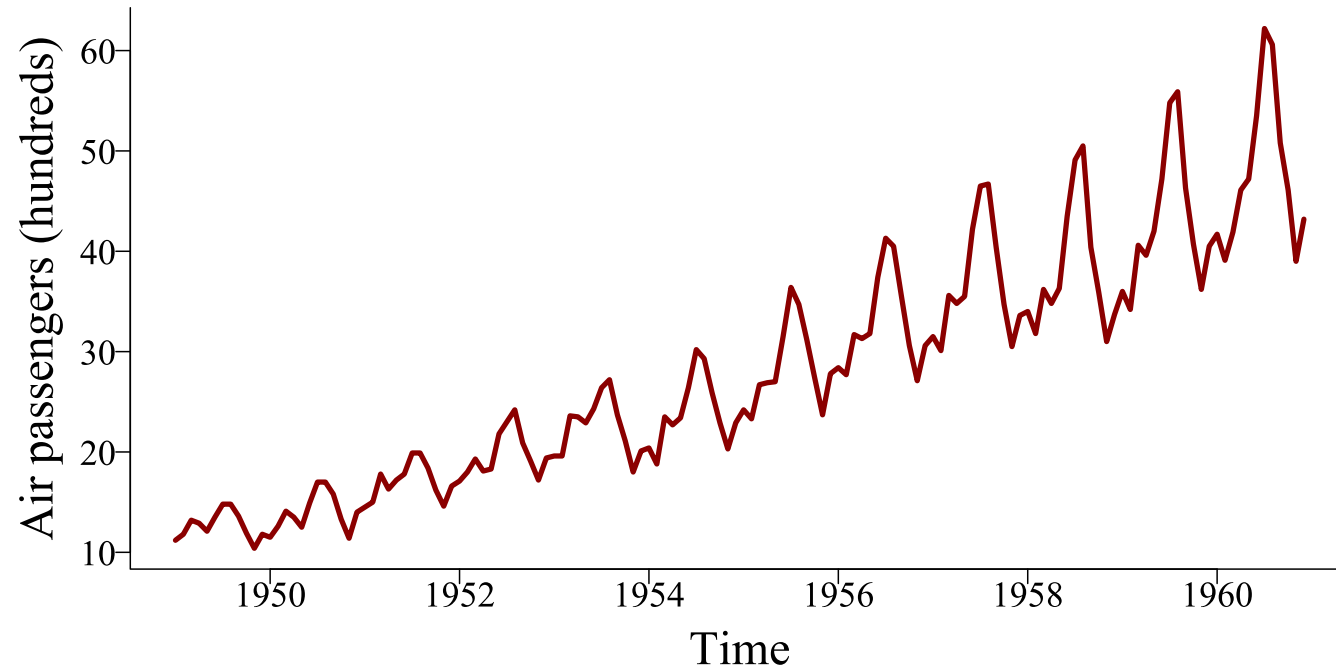
Lunar cycles

Predator / prey dynamics

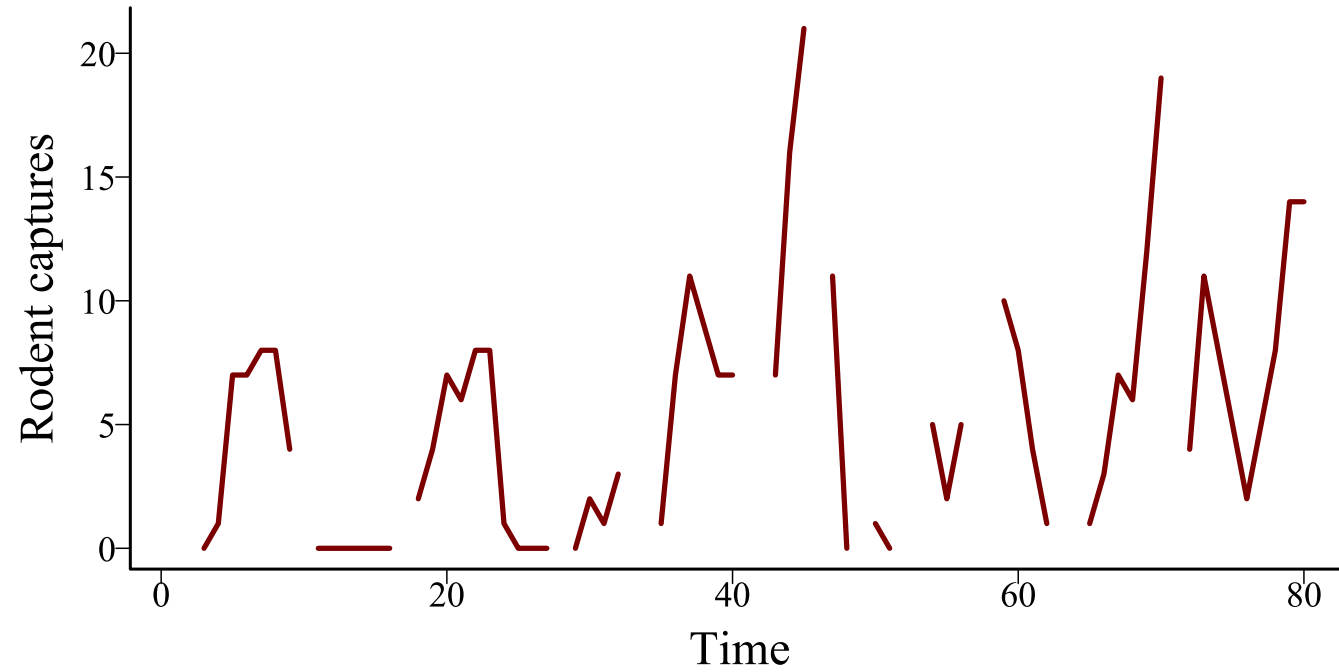
Often change slowly over time



Example seasonal series



Another seasonal series



Visualizing time series

Detecting lagged effects

Lag plots

Autocorrelation functions (ACFs)

Partial autocorrelation functions (pACFs)

Independent correlations

Code Plot

```
# set seed for reproducibility
set.seed(1111)

# number of timepoints
T ← 100

# use arima.sim to simulate from an AR(1) model
series ← arima.sim(model = list(ar = c(0.8)), n = T, sd = 1)

# plot the empirical ACF
acf(series, lwd = 2, bty = 'l',
     ci.col = 'darkred', main = '')
```

Conditional correlations

Code Plot

```
# set seed for reproducibility
set.seed(1111)

# number of timepoints
T ← 100

# use arima.sim to simulate from an AR(1) model
series ← arima.sim(model = list(ar = c(0.8)), n = T, sd = 1)

# plot the empirical pACF
pacf(series, lwd = 2, bty = 'l',
      ci.col = 'darkred', main = '')
```

ACFs often detect seasonality

Code Plot

```
# load the 'gas' dataset from the forecast library  
library(forecast)  
data(gas)  
  
# subset to the final 100 observations  
gas ← gas[377:476]  
  
# plot the empirical ACF over 48 lags  
acf(gas, lag.max = 48, lwd = 2, bty = 'l',  
    ci.col = 'darkred', main = '')
```

But why did we subset?

```
# load the 'gas' dataset from the forecast library
```

```
library(forecast)
```

```
data(gas)
```

```
# subset to the final 100 observations
```

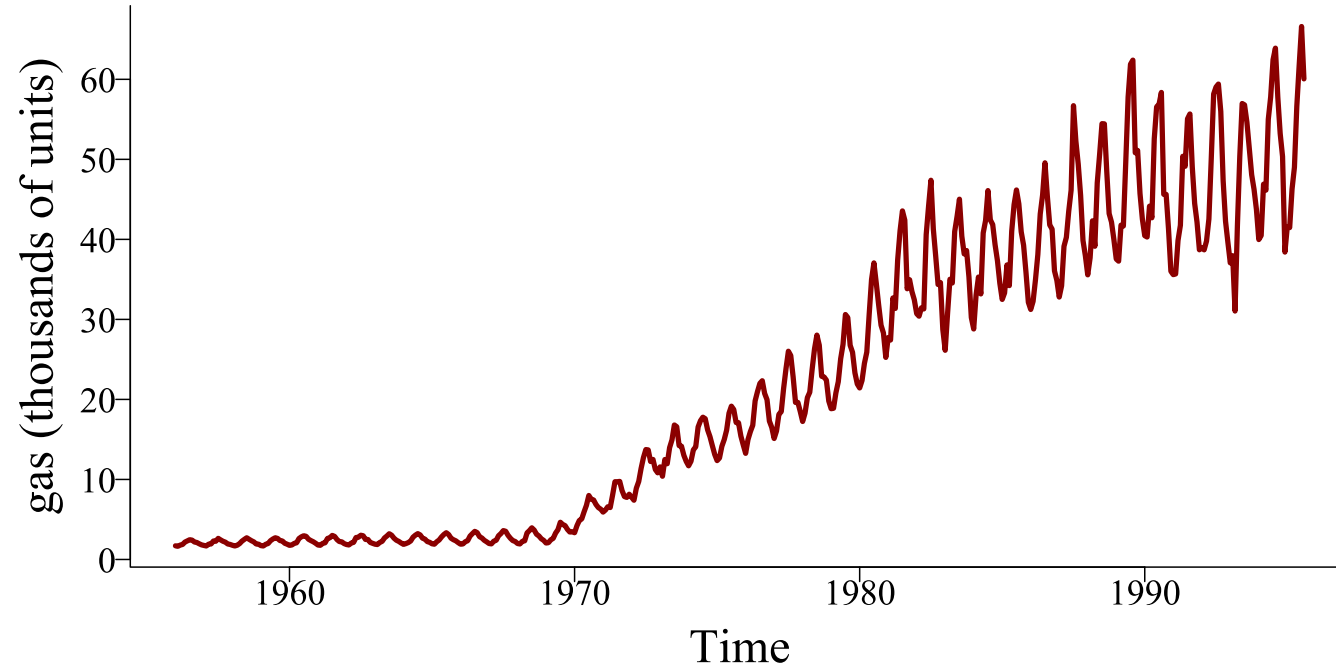
```
gas ← gas[377:476]
```

```
# plot the empirical ACF over 48 lags
```

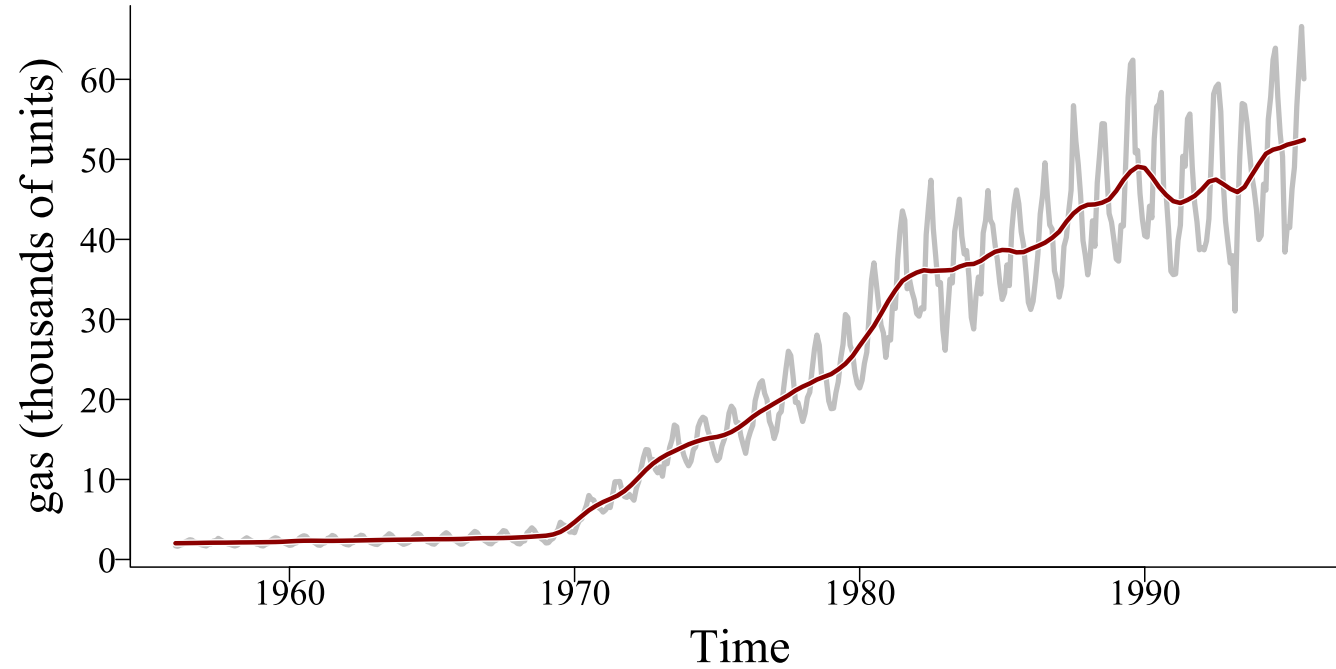
```
acf(gas, lag.max = 48, lwd = 2, bty = 'l',
```

```
    ci.col = 'darkred', main = '')
```

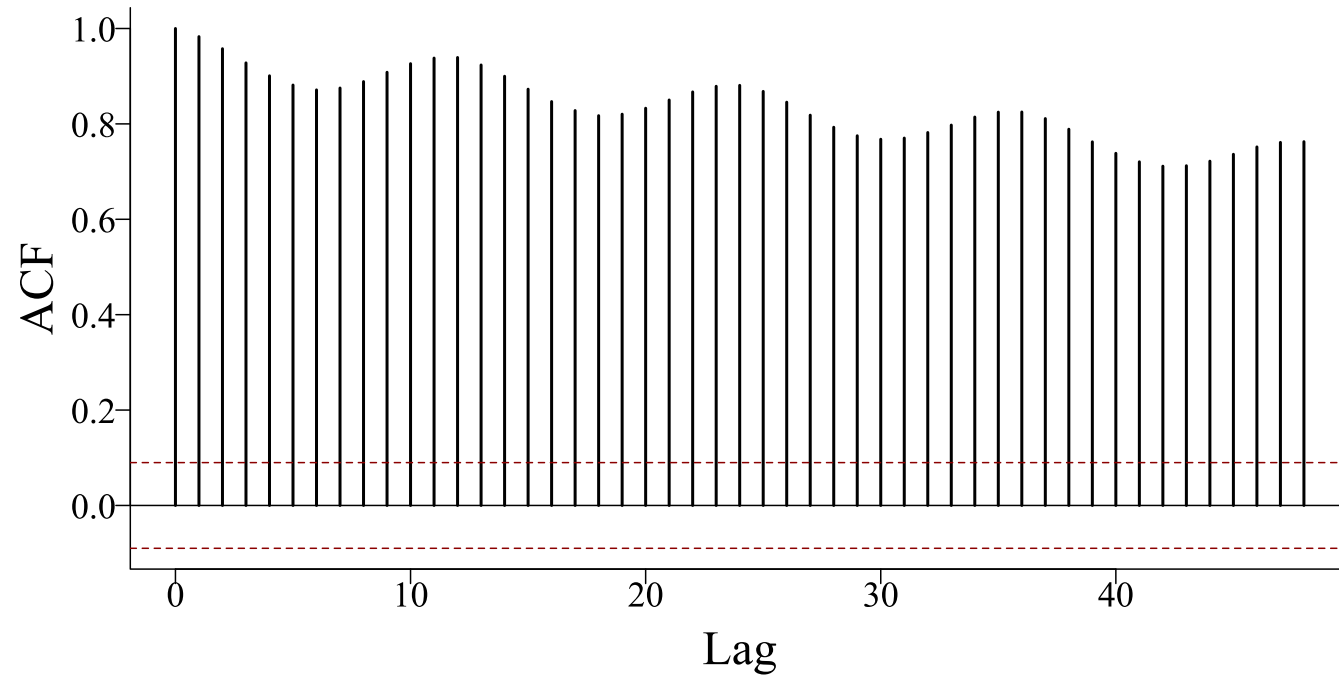
Because gas looks like this ...



... and has a nonlinear *trend*



Raw ACF is misleading



Decompositions

Often it is helpful to split (i.e. decompose) a time series into several sub-components

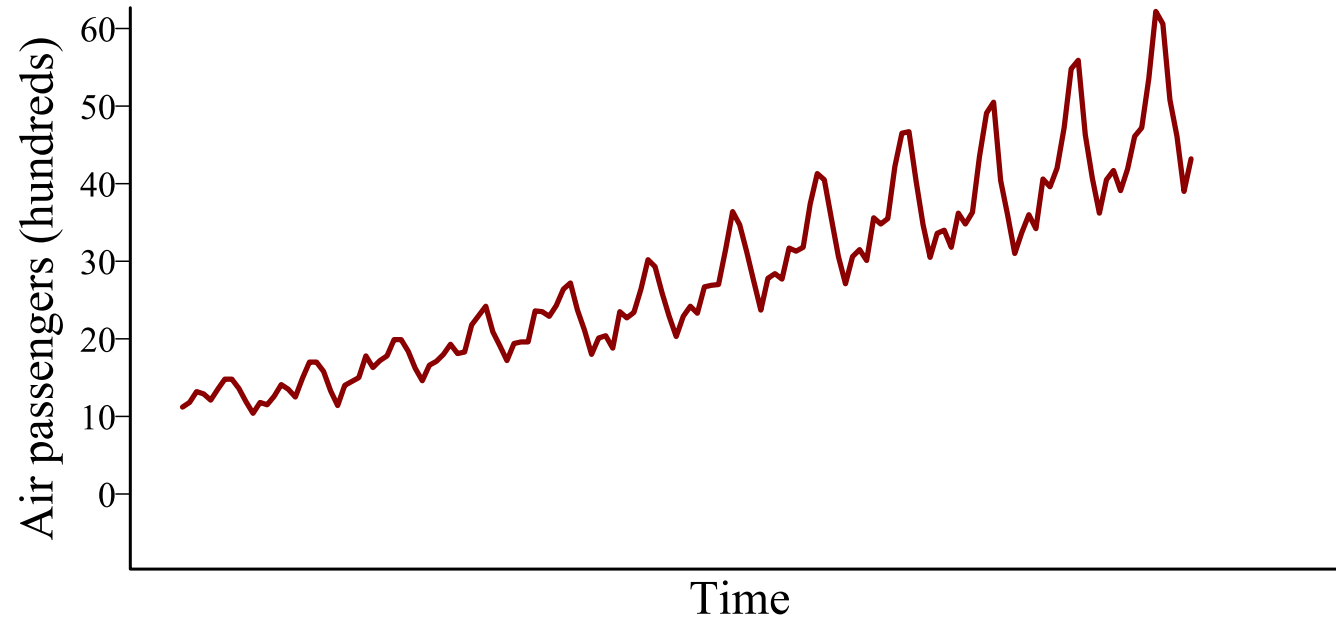
- Long-term trends

- Repeated seasonal patterns

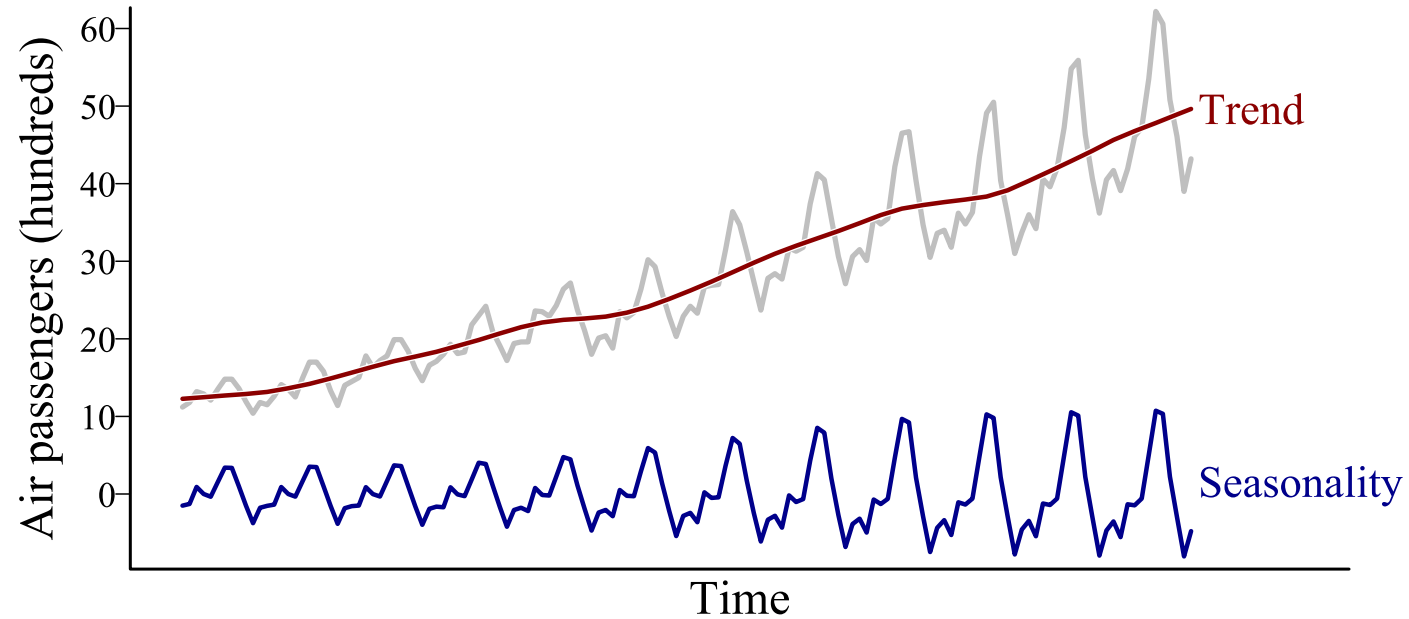
- Remaining non-temporal variation

These components can be summed to give the original series

Example: a complex series



Decompose: trend + seasonality



Modelling these multiple components, either additively or multiplicatively, is a major goal of most time series analysis procedures

Common time series models

Common time series models

Random Walk (RW)

Autoregressive (AR)

Autoregressive Integrated Moving Average (ARIMA; require stationarity)

Exponential Smoothing (ETS)

Regression with ARIMA errors

Very easy to apply in



Hyndman's tools in the forecast  are hugely popular and accessible for time series analysis / forecasting

ETS handles many types of seasonality and nonlinear trends

Regression with ARIMA errors includes additive fixed effects of predictors while capturing trends and seasonality

Some of these algorithms can handle missing data

All are extremely fast to fit and forecast

Great! But what about these?

Temporal autocorrelation

Lagged effects

Non-Gaussian data and missing observations

Measurement error

Time-varying effects

Nonlinearities

Multi-series clustering

Time series
models fail in
ecology

Ecological time series include

Counts of multiple species over time

Presence-absence of species

Repeated captures in multiple plots

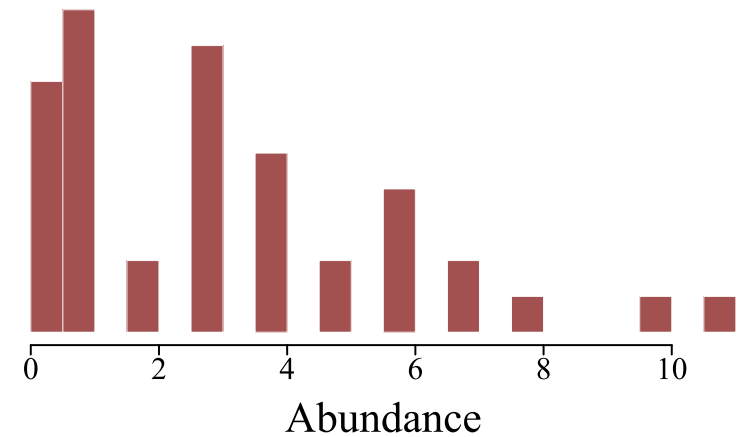
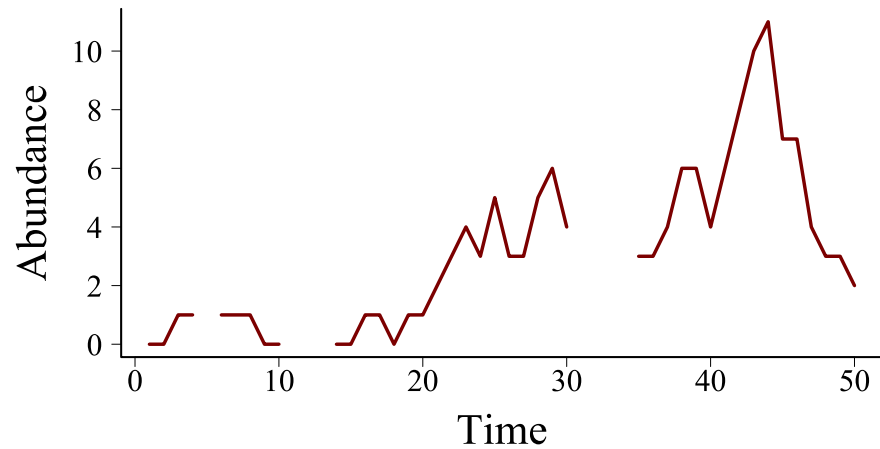
Censored measures (OTUs / pollutants with limits of detection)

Phenology records

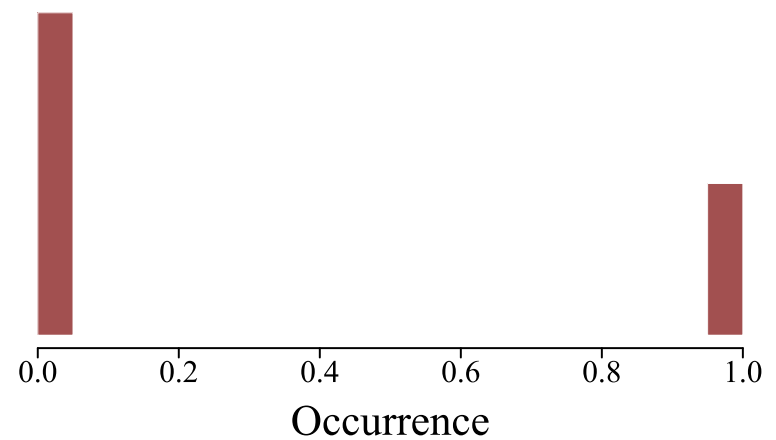
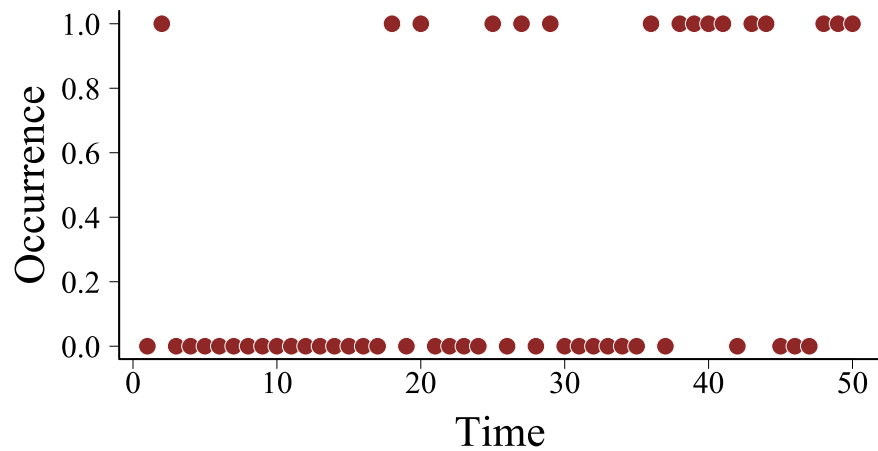
Tree rings

etc...

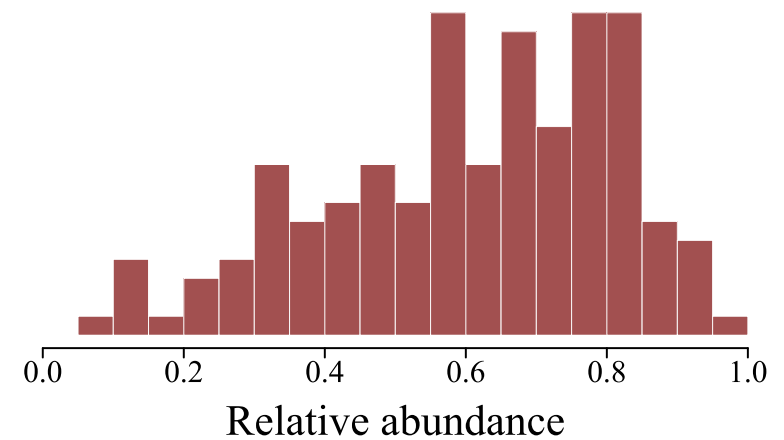
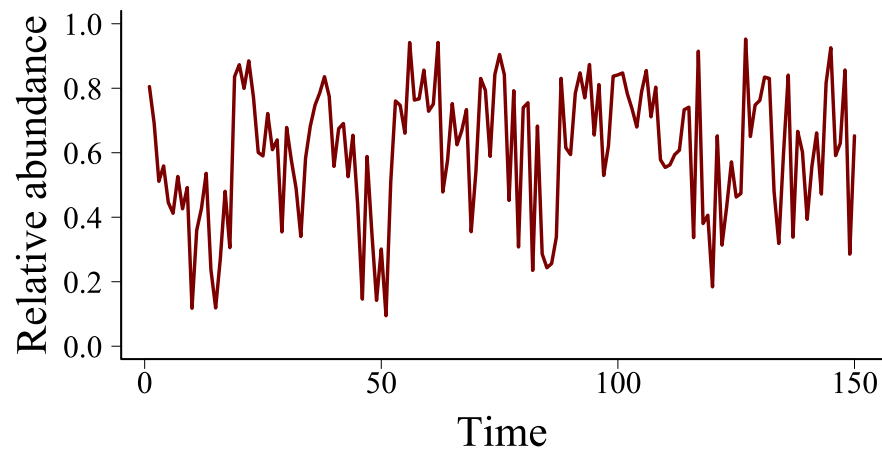
Example ecological time series



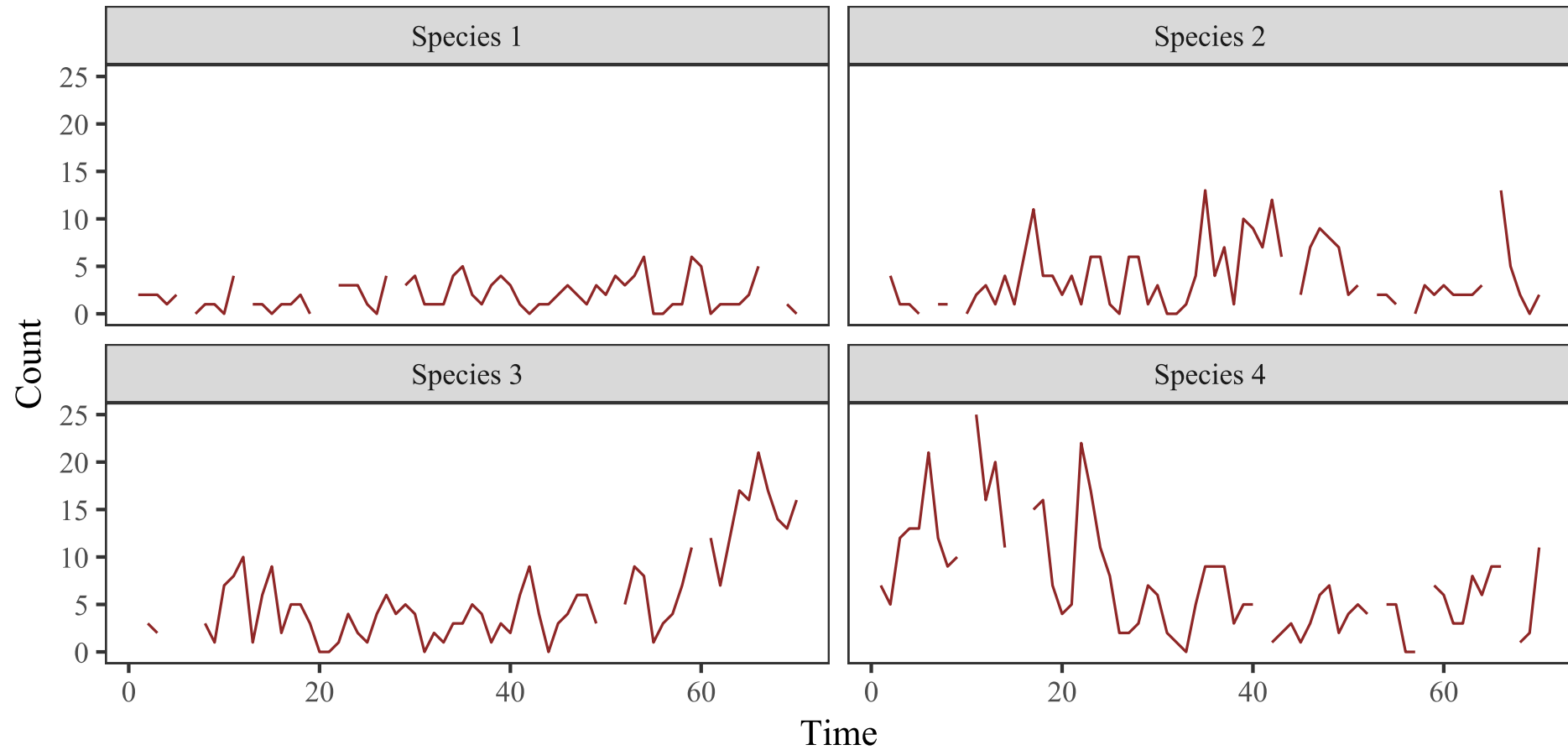
Another ecological time series



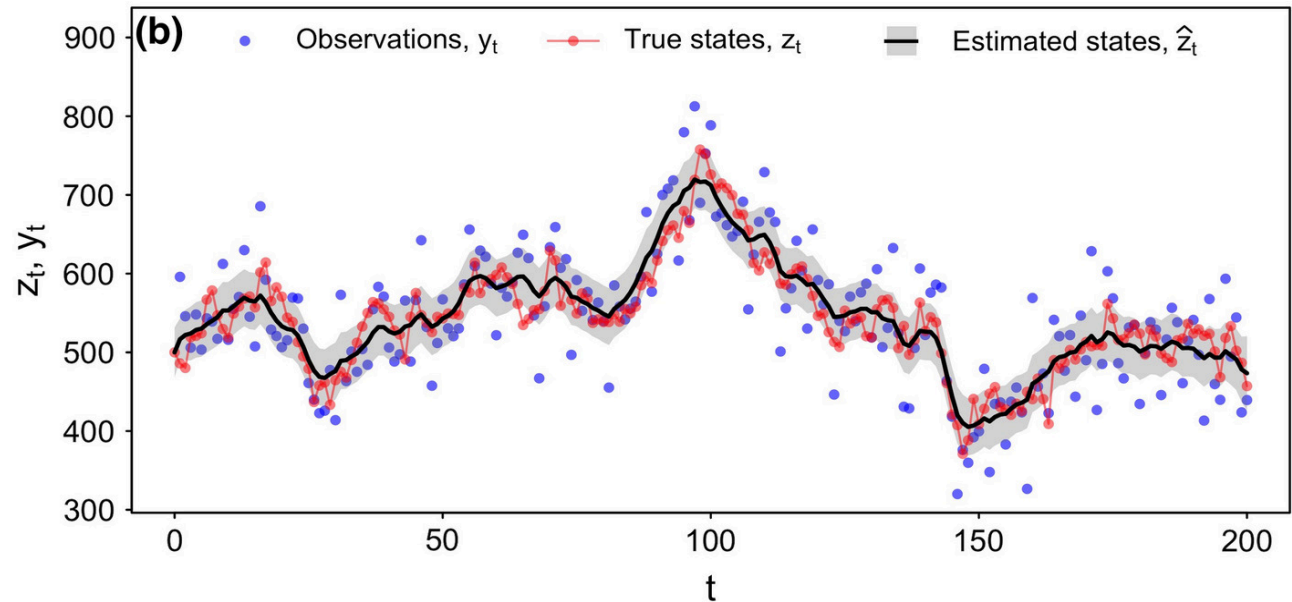
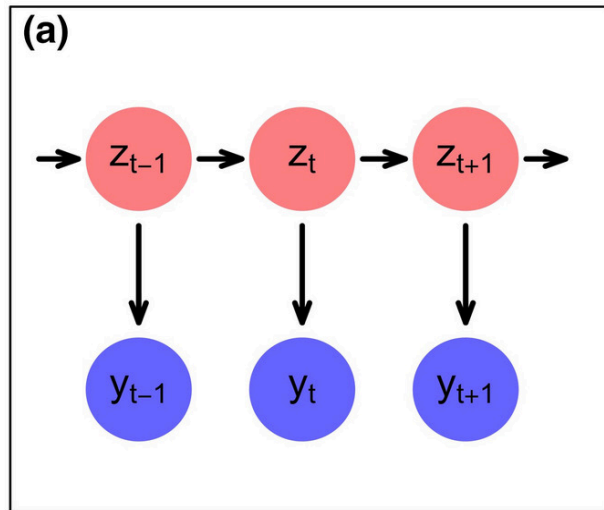
Yet another ecological time series



Collections of ecological series



All can have measurement error



Auger-Methe *et al* 2021

How can we
do better?

In the next lecture, we will cover

Useful probability distributions for ecologists

Generalized Linear and Additive Models

Temporal random effects

Temporal residual correlation structures